

25.12.23

Physics

Ans 1. (3) concave mirror

Ans 2. $u = -10 \text{ cm}$

$$m = -3$$

$$m = \frac{v}{u}$$

$$-3 = \frac{v}{-10}$$

$$v = 30 \text{ cm} //$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{30} + \left(\frac{1}{-10}\right) = \frac{1}{f}$$

$$\frac{1}{f} = \frac{1+3}{30}$$

$$\frac{1}{f} = \frac{4}{30}$$

$$f = \frac{30}{4}$$

$$f = 7.5 \text{ cm} //$$

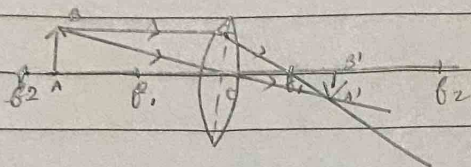
(3) 30 cm, 7.5 cm

Ans 3. (4) Both 2 and 3

Ans 4. (i) Focus :- After reflection where the points converge in a concave mirror and appear to diverge from in a convex mirror to form an image is called the focus or principal focus as it lies on the principal axis, It is denoted by 'F'.

(ii) Centre of curvature :- The centre of curvature is the centre of the sphere of which the spherical mirror is a part. It is denoted by 'C'.

Ans 5.



$$v = +18$$

$$m = \frac{h_i}{h_o} = \frac{v}{u} = \frac{-1}{5}$$

$$\frac{18}{u} = \frac{-1}{5}$$

$$18 \times 5 = -u$$

$$u = -90$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{18} + \frac{1}{90}$$

$$\frac{1}{f} = \frac{5+1}{90}$$

$$f = \frac{90}{6}$$

$$f = 15 \text{ cm}$$

The nature of the lens is a convex lens, Converging lens.

Ans 6. (i) $h_o = 7 \text{ cm}$

$$u = -27 \text{ cm}$$

$$f = -18 \text{ cm}$$

By Mirror formula,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{-18} - \frac{1}{-27}$$

$$\frac{1}{v} = \frac{-3+2}{54}$$

$$\frac{1}{v} = \frac{-1}{54}$$

$$v = -54 \text{ cm}$$

$$m = \frac{h_i}{h_o} = \frac{-v}{u}$$

$$\frac{h_i}{7} = \frac{-(-54)}{-27}$$

$$h_i = \frac{54 \times 7}{-27}$$

$$h_i = -14$$

$$m = \frac{-14}{7}$$

$$m = -2$$

∴ The image is magnified and its a real image.
and the screen should be placed 54 cm
in front of the mirror to get the image.

$$(ii) R = 5 \text{ cm}$$

$$u = -20 \text{ cm}$$

$$f = \frac{R}{2} = \frac{5}{2} = 2.5 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{f} - \frac{1}{u} = \frac{1}{v}$$

$$\frac{1}{v} = \frac{1}{2.5} + \frac{1}{20}$$

$$\frac{1}{v} = \frac{10}{25} + \frac{1}{20}$$

$$\frac{1}{v} = \frac{40+5}{100}$$

$$v = \frac{100}{45}$$

$$v = 2.22 \text{ cm}$$

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

$$= \frac{-2.22}{+20}$$

$$= -0.11$$

∴ The image is smaller than the object, the image will be smaller, and the nature of image is virtual, upright and diminished.

(iii) The mirror used by the dentist is Concave mirror. This is because concave mirror forms a virtual, erect and enlarged image when the object is placed within focus.

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